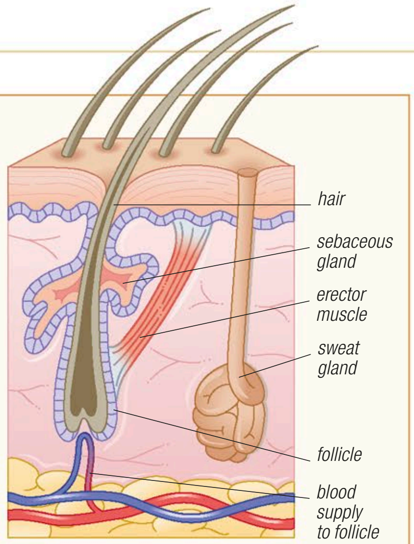
HAIR

Mammals are the only animals with a covering of hair on the body. A hair consists of a rod of cells strengthened by the protein keratin. Hair can take several forms, including whiskers, spines, prickles, and even horns (as in rhinoceroses). Most common is fur, which usually consists of an insulating layer of underfur and a projecting guard coat that protects the skin and gives hair its color (which may aid camouflage). Hairs such as whiskers may also have a sensory function.



SKIN SECTION
Each hair arises from a pocket in the skin called the follicle. Next to the follicle is an erector muscle that raises or lowers the hair, changing the insulating properties of the coat.

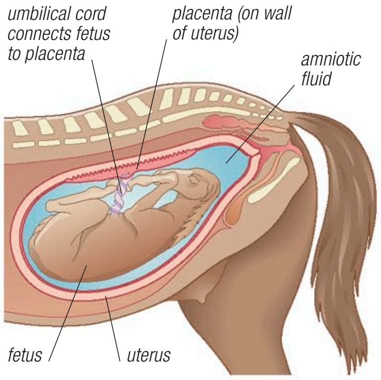
MODIFIED HAIR
Porcupines have whiskers, long body hairs, and defensive spines (which are modified hairs).

they are transmitted to the inner ear by 3 tiny bones, and so to the brain. The fennec fox has enormous, sensitive pinnae; by contrast, true seals have lost their pinnae.

One of the most distinctive parts of a mammal's body is its skin. This consists of 2 layers: a protective outer layer of dead cells (the epidermis) and an inner layer (the dermis) that contains blood vessels, nerve-endings, and glands. It is the glands in the dermis that are particularly unusual: the sebaceous (or scent) glands secrete chemicals that mammals use to communicate with one another, the mammary glands produce the milk used to nourish newborn young, and the sweat glands—together with the hair, which also arises from the dermis (see panel, above)—play an important part in regulating temperature.

Reproduction

Depending on the way they reproduce, mammals are divided into 2 groups or subclasses—Prototheria and Theria—that is, the egg-layers and those that give birth to live young. Theria is subdivided into 2 further groups—infraclass Metatheria (the marsupials) and infraclass Eutheria (other mammals). Marsupials complete their development while being nourished by their mother's milk, whereas placental mammals develop within the body of the mother and nourishment is supplied to the fetus through the placenta. Marsupials have no true placenta (see below) and produce young at a very early stage of their development. In some species, the young are kept in a pouch on the outside of the mother's body until they are more fully grown. The largest reproductive group is the infraclass Eutheria, which contains the placental mammals. The unborn young develop in the mother's uterus. During pregnancy, food and oxygen pass from mother to fetus through an organ known as the placenta, while waste substances move in the opposite direction. When born, infant placental mammals are more highly developed than those of marsupials.



PLACENTAL MAMMALS
The fertilized egg of a placental mammal, such as a horse, divides many times and eventually becomes a fetus. As the fetus grows, the uterus expands in size and weight. In the horse, pregnancy lasts about 11 months.

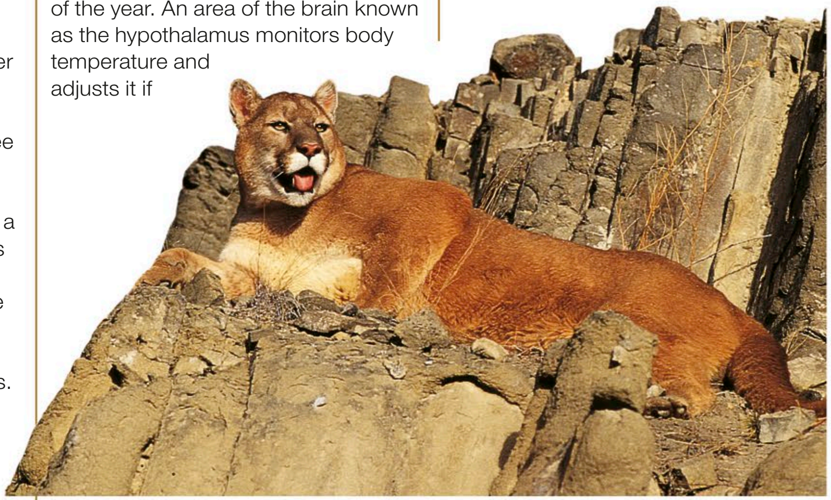
The juveniles of all mammals are fed on milk secreted by the mother's mammary glands, which become active after the young are born. Except in monotremes, the milk is delivered through teats. As well as providing nourishment (it is rich in proteins and fats), the milk contains antibodies that help establish resistance to infection. Being nourished on milk during the early weeks of life also means that young mammals do not have to forage for their own food, which greatly increases their chances of survival.

Litter sizes vary from 20 (Virginia opossum) to only 1 (orangutan); and gestation periods range from 12 days (short-nosed bandicoot) to 22 months (African savanna elephant).

Temperature control

Mammals, like birds, are endothermic, meaning that they maintain a constant body temperature and can therefore remain active at extremely high or low external temperatures. This is why mammals are able to occupy every major habitat and are more widespread than any other vertebrates (except birds). Many species, such as the seals and whales of the Antarctic, live in regions where the temperature is well below freezing for much or all of the year. An area of the brain known as the hypothalamus monitors body temperature and adjusts it if necessary. Body temperature can be altered by increasing or decreasing the metabolic rate, widening or constricting blood vessels that carry heat to the skin's surface, raising or lowering body hair to trap or release an insulating layer of air, and causing shivering (to gain heat) or evaporation via sweating or panting (to lose heat). Mammals can also control their body temperature by adopting special body postures: a monkey, for example, will hunch up in the cold (and many mammals huddle together in small groups to keep warm); lemurs warm up in the early morning sun by sitting up and

STAYING WARM
Large animals, such as red deer, remain active all winter. They survive and keep warm by using fat reserves built up during the summer. Over time, this leads to weight loss.



KEEPING COOL
In hot climates, mammals avoid overheating by resting in the hottest part of the day. They may also keep cool by panting, as this puma is doing. Panting helps lower the body temperature through the evaporation of water from internal surfaces, such as the tongue.

HIBERNATION

Some mammals, especially small species, conserve energy during the cold months by hibernating, just as some reptiles (for example, the garter snake) do. Their temperature and heart rate fall, their breathing slows, their metabolism drops to almost imperceptible levels, and they fast, drawing on stored fat reserves. When hibernating, the animal is difficult to rouse. A West European hedgehog begins hibernation when the outside temperature falls below 59° F (15° C), and in midwinter its body temperature lowers to about 43° F (6° C). In some bats, rectal temperatures of 32° F (0° C) have been recorded during hibernation. Larger animals, such as the American black bear, do not truly hibernate—they sleep. Their body temperature drops only a few degrees and they can rouse more easily. Related to hibernation is estivation, which is torpidity during the summer. Like hibernation, estivation saves energy when food is short.

COLD SLEEP
Many bats that live in temperate regions hibernate over the winter. While sleeping, the temperature of their body falls to that of the roost site, as shown by the dew on this Daubenton's bat, right.

WINTER HIDEAWAY
For hedgehogs, effective camouflage is an important part of winter hibernation.